EXPERIMENT 7 – IMPLEMENTATION OF THE SLIDING WINDOW PROTOCOL

AIM

To implement flow control at the data link layer using the Sliding Window Protocol and simulate the transfer of frames from a sender to a receiver.

THEORY

The Sliding Window Protocol is a data-link layer method for controlling the rate of frame transmission. It allows a sender to send several frames before it must wait for acknowledgements from the receiver.

In this experiment, every character in the input message is treated as an individual frame containing a frame number and its data. The sender sends frames in groups according to the selected window size.

The receiver checks whether each incoming frame number matches the expected sequence. It sends an ACK when the frame is accepted and a NACK when an unexpected frame is detected.

The sender checks the acknowledgement information. On receiving the expected ACKs, it advances the window and sends the next group of frames. If a NACK occurs, the necessary frames are sent again.

The experiment also demonstrates how modifying a frame number or acknowledgement in the buffer files can be used to simulate an error.

ALGORITHM

Sender

1. Begin the program.
2. Get the window size from the user.
3. Read the message to be transmitted.
4. Treat each character as a separate frame.
5. Assign a sequential number to each frame.
6. Prepare frames containing the frame number and character.
7. Transmit a group of frames according to the window size.
8. Write the transmitted frames to `Sender_Buffer`.
9. Wait for acknowledgement information from the receiver.
10. Read the contents of `Receiver_Buffer`.
11. Verify the received ACK or NACK values.

12. If the ACKs are valid, slide the window forward.

13. Send the following group of frames.

14. If a NACK is received, retransmit the required frames.

15. Repeat until every frame has been sent successfully.

16. End the program.

Receiver

1. Start the receiver process.

2. Read the transmitted frames from `Sender_Buffer`.

3. Examine the frame number in each received frame.

4. Compare the number with the next expected frame number.

5. If the sequence is correct, accept the frame and generate an ACK.

6. If the sequence is incorrect, generate a NACK.

7. Save acknowledgement details in `Receiver_Buffer`.

8. Continue until all frames have been processed.

9. Stop the receiver process.

PROGRAM

The Python program accepts a message and window size, creates numbered frames, writes each transmitted window to `Sender_Buffer`, and records ACK/NACK responses in `Receiver_Buffer`. It then advances the window when acknowledgements are successful.

```
import time

message = input("Enter the message: ")
window = int(input("Enter window size: "))

frames = []
for i, ch in enumerate(message):
    frames.append([i + 1, ch])

i = 0
while i < len(frames):
    end = min(i + window, len(frames))
    print("\nSending frames:")

    with open("Sender_Buffer", "w") as f:
        for j in range(i, end):
            print("Frame No:", frames[j][0], "Data:", frames[j][1])
            f.write(str(frames[j][0]) + " " + frames[j][1] + "\n")

    time.sleep(1)
    expected = i + 1

    with open("Receiver_Buffer", "w") as f:
        for j in range(i, end):
```

```
            if frames[j][0] == expected:
                f.write("ACK " + str(frames[j][0]) + "\n")
                print("Receiver: ACK", frames[j][0])
                expected += 1
            else:
                f.write("NACK " + str(expected) + "\n")
                print("Receiver: NACK", expected)
                break

    with open("Receiver_Buffer", "r") as f:
        ack = f.readlines()

    success = True
    for line in ack:
        parts = line.split()
        if parts[0] == "NACK":
            success = False
            break

    if success:
        i = end
    else:
        print("Retransmitting frames...")
        time.sleep(1)

print("\nAll frames transmitted successfully.")
print("Message:", message)
```

SAMPLE INPUT

Enter the message: HELLO

Enter window size: 3

SAMPLE OUTPUT

Sending frames:

Frame No: 1 Data: H

Frame No: 2 Data: E

Frame No: 3 Data: L

Receiver: ACK 1

Receiver: ACK 2

Receiver: ACK 3

Sending frames:

Frame No: 4 Data: L

Frame No: 5 Data: O

Receiver: ACK 4

Receiver: ACK 5

All frames transmitted successfully.

Message: HELLO

OBSERVATION

Frame No.	Data	Acknowledgement
1	H	ACK 1
2	E	ACK 2
3	L	ACK 3
4	L	ACK 4
5	O	ACK 5

For a window size of 3, the sender first transmits three frames. After the acknowledgements arrive, the window advances and the remaining frames are sent. If a wrong frame number is introduced, the receiver can send a NACK and request retransmission.

RESULT

Thus, the Sliding Window Protocol was implemented to demonstrate data-link flow control. Frame transfer between sender and receiver was simulated using acknowledgements and retransmission handling.